



HABIB UNIVERSITY

Data Structures & Algorithms

CS/CE 102/171 Spring 2025

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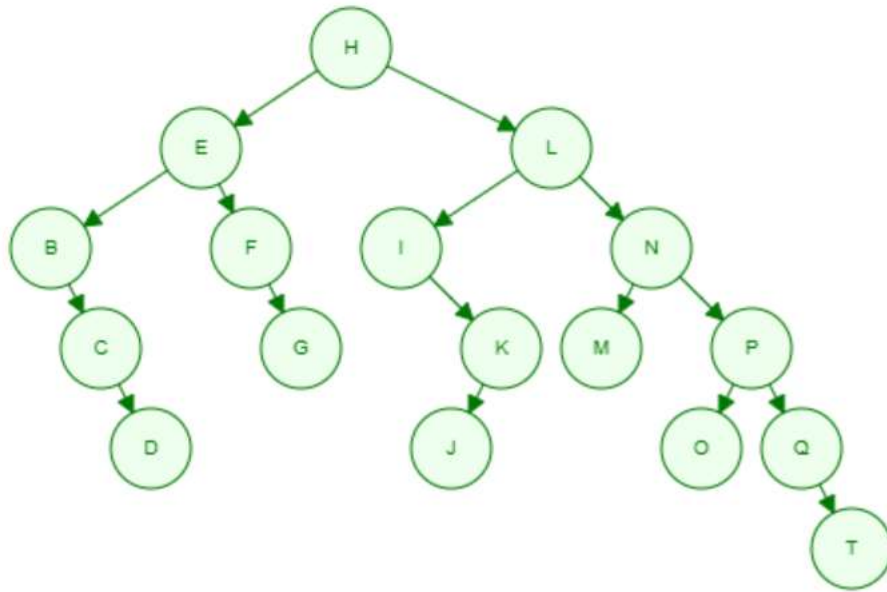
Quiz # 5 Solution

QUESTION: Binary Search Tree

1. For the given set of nodes, construct a binary search tree:

- Nodes = {H, L, I, E, N, B, C, K, P, D, F, G, O, Q, J, M, T}

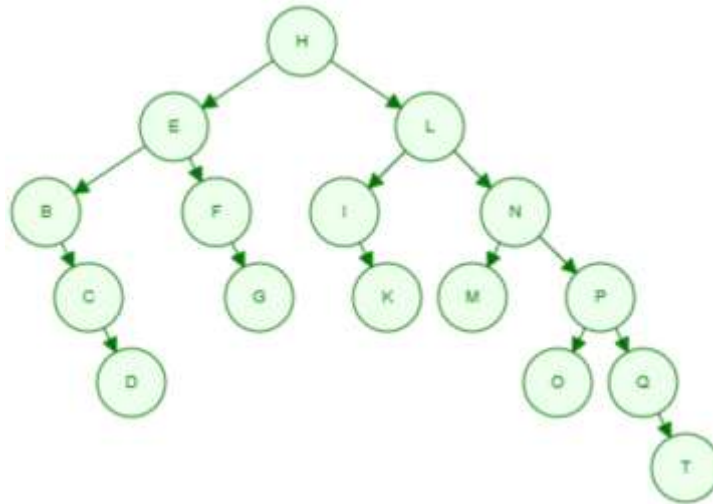
Resultant BST:



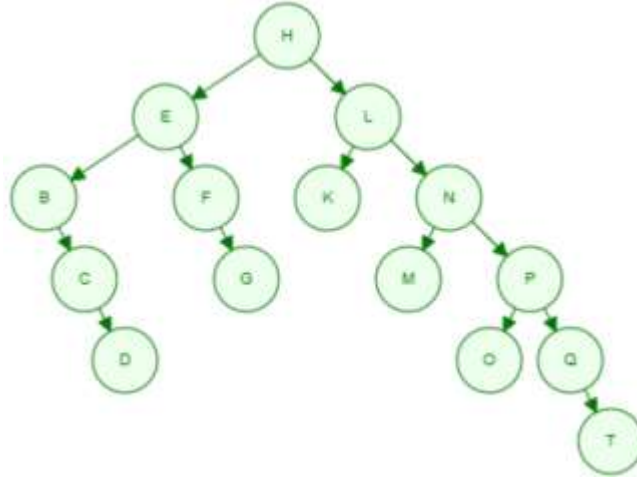
2. Search for Node O in the BST constructed in part (1), by specifying its complete path including its directions:
- Root: H → Right: L → Right: N → Right: P → Left: O**
Therefore, successfully found!
3. Search for the minimum in the BST constructed in part (1), by specifying its complete path including its directions:
- Root: H → Left: E → Left: B → Left: Null**
Therefore, minimum = Node B
4. Find the successor of Node G in the BST constructed in part (1), by specifying its complete path including its directions:
- Node: G → Parent: F → Parent: E → Parent: H**
 - Node H is immediately bigger value than Node G, so that is its inorder successor**
5. Find the predecessor of Node L in the BST constructed in part (1), by specifying its complete path including its directions:
- Node: L → Left: I → Right: K → Right: Null**
Therefore, predecessor of Node L is Node K

6. Delete the following nodes one after the other from the BST constructed in part (1). Redraw the tree to get complete grade for deletion steps. Use inorder successor wherever needed for deletion.

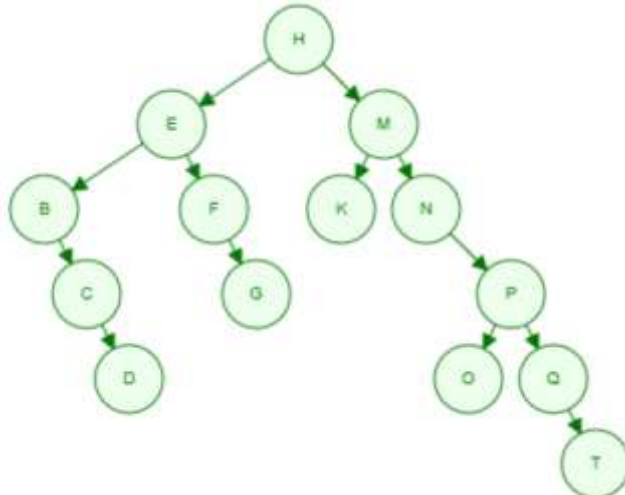
- Node J: **Node J is a leaf node, so it can directly be removed from the tree, to get the resultant BST as follows:**



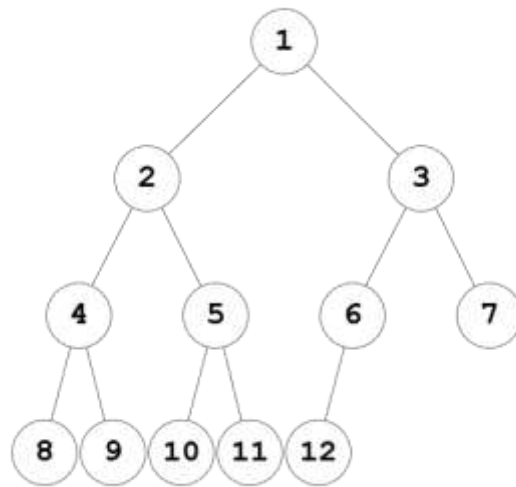
- Node I: **Node I has one child, so connect the child to I's parent, and get following BST:**



- Node L: **Node L has two children. According to definition, we can replace it with its inorder successor or inorder predecessor, but as the question explicitly asks to replace with its successor, we will find its successor first**
 - Node: L → Right: N → Left: M → Left: Null**
 - Hence, M is the successor of L, so replace L with it to get following resultant BST:**



7. For the tree given below, write down its Linked Structure Representation:



- **{ Root : 1 ,
Left : {
 Root : 2,
 Left : {
 Root : 4,
 Left : 8,
 Right : 9
 },
 Right : {
 Root : 5,
 Left : 10,
 Right : 11
 }
},
Right : {
 Root : 3,
 Left : {
 Root : 6,
 Left : 12,
 Right : None
 },
 Right : 7
}
}**